1. **Course number and name**
   CHE 321. Chemical Reaction Engineering

2. **Credits, contact hours and categorization**
   Credit hours: 3; Contact hours: 3 lecture hours per week; Engineering

3. **Instructor's or course coordinator's name**
   Korosh Torabi, Meenesh Singh

4. **Textbook, title, author, publisher, year**

"Elements of Chemical Reaction Engineering",
Fifth, Fourth or Third Edition, by Fogler, H. Scott

5. **Specific course information**
   a. Brief description of the content of the course (catalog description)
      Kinetics of homogeneous single reactions. Ideal reactors: batch, stirred tank and plug flow systems. Conversion and yield in multiple reactions. Design and optimization of reactors: non-isothermal, catalytic, and non-ideal reactors.
   b. Prerequisites or co-requisites
      CHE 210, CHE 301, MATH 220.
   c. Indicate whether a required, elective, or selected elective.
      Required.

6. **Specific goals for the course**
   a. Specific outcomes of instruction.

1. Define the rate of chemical reaction. Apply the mole balance equations to a batch reactor, continuous stirred tank reactor, plug flow reactor, and packed bed reactor.
2. Define conversion and space time. Write the mole balances in terms of conversion. Determine reactor sizes (volume, catalyst weight).
3. Understand rate laws and the Arrhenius equation. Describe homogeneous, heterogeneous, elementary, nonelementary and reversible reactions.
4. Apply reaction stoichiometry to molar and volumetric flow rates of a species and conversion.
5. Describe the CRE algorithm to solve chemical reaction engineering problems. Apply this algorithm to design isothermal reactors.
6. Determine reaction order and rates from experimental data of batch or flow reactors.
7. Understand integral and differential methods to identify rate parameters.
8. Describe yield and selectivity. Apply CRE algorithm to design the reactor with multiple reactions for maximal selectivity.
9. Modify CRE algorithm to design non-isothermal reactors (adiabatic and non-adiabatic) and optimize the reactor staging.
10. Analyze multiple steady states and optimal operation of reactors.
11. Define a catalyst, derive a catalytic reaction mechanism and rate-limiting step
12. Obtain rate expressions from quasi-equilibrium hypothesis.
13. Understand catalyst deactivation mechanism Analyze heterogeneous catalytic reactors
14. Define residence time distribution (RTD), RTD functions

15. Evaluate non-ideal reactor models.

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome		1	2	3	4
(1)	an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				X
(2)	an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.				X
(3)	an ability to communicate effectively with a range of audiences.		X		
(4)	an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.			X	
(5)	an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.			X	
(6)	an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.		X		
(7)	an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.			X	

7. Brief list of topics to be covered

Reaction fundamentals
Reactor Types, General Mole Balances, Conversion, Reactor Sizing, Stoichiometric Tables
Isothermal Reactor Design
SOFTWARE - MATLAB for Reactor Design
Pressure Drop in Reactors
Unsteady State Operation, Rate Analysis
Multiple Reactions
General Energy Balance & CSTR, PFR Energy Balance, Batch Energy Balance
Equilibrium Conversion
Non-Isothermal Reactor Example
Multiple Steady States in a CSTR/Multiple Steady States/Multiple Reactions
Summary of Non-Isothermal Reactors
Catalytic Reactions, Langmuir-Hinshelwood Rate Laws
Langmuir-Hinshelwood Rate Laws, Heterogeneous Reactors, Mass Transfer
External and Internal Mass Transfer Limitations
Effectiveness Factor and Falsified Kinetics
Residence Time Distribution
Zero-Parameter Flow Models, Non-Ideal Flow and combination of ideal reactors